4. Lessons from the History of Interpretability

Causal interpretability techniques have existed since the beginning of deep learning. What distinguishes the current wave of mechanistic interpretability studies from past causal interpretability work? What actionable lessons can past work (which often used very different methods and mediators to contemporary studies) teach us about analyzing intermediate model computations? We claim that the lens of causal mediation analysis (1) enables a novel and clear narrative of the trajectory of interpretability research; (2) links current issues in the field to longstanding issues that have existed since at least the 1980s; and (3) highlights actionable research directions. We focus primarily on (1) and (2) in this section, and return to (3) in §7.

Interpretability at the beginning of deep learning. In 1986, Rumelhart, Hinton, and Williams published an algorithm for neural network backpropagation and an analysis of this algorithm. This enabled and massively popularized research into multi-layer perceptrons (MLPs)—now often called feedforward layers. That work arguably represents the first mechanistic interpretability study: the authors evaluated their method by inspecting each activation and weight in the neural network, and observing whether the learned algorithm corresponded to the human intuition of how the task should be performed. In other words, they reverse-engineered the algorithm of the network by labeling the rules encoded by each neuron and weight!

From the 1980s through the early 2000s, rule extraction via neuron-level activation and weight analysis remained popular. At first, this was a manual process: networks were either small enough to be manually interpreted (Rumelhart, Hinton, and Williams 1986; McClelland and Rumelhart 1985) or interpreted with the aid of carefully crafted datasets (Elman 1989, 1990, 1991). For example, Elman (1990, 1991) found that recurrent neural networks were capable of capturing hierarchical semantic relationships, and were sensitive to syntactic context. Alternatively, researchers could prune the network (Mozer and Smolensky 1988; Karnin 1990) to a sufficiently small size to be manually interpretable. Later, researchers proposed techniques for automatically extracting rules (Hayashi 1990) or decision trees from NNs (Craven and Shavlik 1994, 1995; Krishnan, Sivakumar, and Bhattacharya 1999; Boz 2002)—often after the network had been pruned. At this point, interest in automated causal methods based on interventions had not yet been established, as networks were often small and simple enough to directly understand without significant abstraction.

Nonetheless, as the size of neural networks increased, the number of rules that could be encoded in a network increased. Thus, rule/decision tree extraction techniques could not generate easily human-interpretable explanations nor algorithmic abstractions of model behaviors beyond a certain size. This led to the rise of **visualization methods** in the 2000s, which became a popular way to demonstrate the complexity of phenomena that models had learned to encode. Visualizations of network inputs and outputs (Tzeng and Ma 2005) and interactive visualizations of model activations (Erhan et al. 2009) were valuable initial tools for generating hypotheses as to what kinds of concepts models could represent. While visualization research was generally *not* causal, this subfield would remain influential for interpretability research as neural networks scaled in size in the following decade.

Large-scale pre-trained models. The 2010s were a time of rapid change in machine learning. In 2012, the first large-scale and widely adopted pre-trained neural network, AlexNet (Krizhevsky, Sutskever, and Hinton 2012), was released. Not long after, pre-trained word embeddings (Mikolov et al. 2013a,b; Pennington, Socher, and Manning 2014) became common in NLP, and further pre-trained deep networks followed (He et al. 2016). These were based on ideas from *deep learning*. This represented a significant paradigm shift: formerly, each study would build ad-hoc models which were not shared across studies, but which were generally more transparent.[11] After 2012, there was a transition toward using a shared collection of significantly larger and more capable—but also more opaque—models. This raised new questions on what was encoded in the representations of these shared scientific artifacts. The rapid scaling of these models rendered old neuron-level rule extraction methods either intractable or made its results difficult to interpret. Thus, interpretability methods in the early 2010s deployed scalable and relatively fast *correlational* methods, including visualizations (Zeiler and Fergus 2014) and saliency maps (Simonyan, Vedaldi, and Zisserman 2014). This trend continued into 2014–2015, when recurrent neural network–based (Elman 1990) language models (Mikolov et al. 2010) began to overtake non-neural statistical models in performance (Bahdanau, Cho, and Bengio 2015); for example, visualizing RNN and LSTM (Hochreiter and Schmidhuber 1997) hidden states was proposed as a way to better understand their incremental processing (Karpathy, Johnson, and Fei-Fei 2016; Strobelt et al. 2017).

At the same time, interpretability methods started to focus more on *explaining* model predictions.[12] The explainable AI (XAI) field was and is extensive. One line of work designed supervised auxiliary (correlational) models to explain particular model predictions, such as LIME (Ribeiro, Singh, and Guestrin 2016b,a), Anchors (Ribeiro, Singh, and Guestrin 2018), and extensions like CLEAR that explicitly integrate notions of counterfactual fidelity to the output explanations (White and d'Avila Garcez 2020). These models learn local decision boundaries, or some human-interpretable simplified representation of a model's behavior. Other works interpreted predictions via feature importance measures like SHAP (Lundberg and Lee 2017). Influence functions (Koh and Liang 2017) traced the model's behavior back to specific instances from the training data. Another line of work sought to directly manipulate intermediate concepts to control model behavior at test time (Koh et al. 2020), or to decompose distributed representations into interpretable symbolic representations post hoc (Odense and Garcez 2020). The primary difference between these visualization-/correlation-/input-based methods and current methods lies in whether they prioritize black-box explanations or white-box explanations—that is, whether they explain model behaviors in terms of input/output relationships or require analysis of model internals, respectively. Black-box explanations allow us to generate hypotheses as to the types of *input concepts* that explain particular model predictions. In contrast, current work prioritizes *white-box explanations*—i.e., highly localized and causal explanations of *how* and in *which components of the computation graph* models perform a given behavior.

2017–2019 featured perhaps the largest architectural shift (among many) in machine learning methods at this time: Transformers (Vaswani et al. 2017) were released and

11 Many systems built before deep learning were based on feature engineering, and so the information they relied on was more transparent than in current systems.

12 There has classically been a distinction between *local* and *global* interpretability; local interpretability is concerned with explaining specific model predictions, whereas global interpretability is concerned with explaining a given model behavior in general across examples (Lipton 2018; Guidotti et al. 2018). Both styles of interpretability can be valuable, depending on one's research question. A benefit of causal mediation analysis is that it can encompass both styles. As recent work has tended to focus more on global interpretability, we devote more attention to this style of work, though we cite and acknowledge examples of local interpretability methods in this section.

quickly became popular due to scalability and high performance. This led directly to the first successful large-scale pretrained language models, such as (Ro)BERT(a) (Devlin et al. 2019; Liu et al. 2019b) and GPT-2 (Radford et al. 2019). These significantly outperformed prior models, but it was unclear *why*—and at this scale, analyzing neural networks at the neuron level using past techniques had become intractable. This combination of high performance and little mechanistic understanding created demand for interpretability techniques that allowed us to see *how* language models had learned to perform so well.

Hence, correlational probing methods rose to meet this demand. In this approach, classifiers are trained on intermediate activations to extract some target phenomenon. Probing classifiers have been used to investigate the latent morphosyntactic structures encoded in static word embeddings (Köhn 2015; Gupta et al. 2015) or intermediate hidden representations in natural language systems—for example, in neural machine translation systems (Shi, Padhi, and Knight 2016; Belinkov et al. 2017; Conneau et al. 2018) and pre-trained language models (Hewitt and Manning 2019; Hewitt et al. 2021; Lakretz et al. 2019, 2021). However, probing classifiers lack consistent baselines, and the claims made in these studies were not often causally verified (Belinkov 2022). For instance, although an intervention may target a causal property of the task $V_i \to Y$, an alternative spurious property V_j may be picked up by the probe, which impedes causal claims about $V_i \to Y$ (Ravichander, Belinkov, and Hovy 2021). This encouraged researchers to search for more causally efficacious methods.

The rise of causal and mechanistic interpretability. 2017–2018 featured the first hints of our current wave of mechanistic interpretability, primarily based on interventions to neurons or full layers. Giulianelli et al. (2018) trained a probing classifier, but then used gradients from the probe to modify the activations of the network. Other studies analyzed the functional role of individual neurons in static word embeddings (Li, Monroe, and Jurafsky 2017) by forcing certain neurons on or off. Parallel developments in computer vision were influential: Bau et al. (2019b) found that interpretable concepts in the outputs of generative adversarial networks (Goodfellow et al. 2014) could be modified via interventions to specific neurons. The idea of manipulating neurons to steer behaviors was then applied to downstream task settings, such as machine translation (Bau et al. 2019a). These techniques were popularized in 2020 when Vig et al. (2020) proposed a method for assigning task-specific causal importance scores to specific neurons and attention heads by systematically computing each component's indirect effect (Eq. 1) on the model's output. It was an application of the counterfactual theory of causality (Lewis 1973, 1986), as well as Judea Pearl's causal mediation analysis framework (Pearl 2001, 2000). This enabled a new line of interpretability research that aimed to faithfully localize model behaviors to specific components—an idea that would become foundational to contemporary causal and mechanistic interpretability.

At the same time, however, researchers began to realize the significant performance improvements that could be gained by massively increasing the number of parameters and training corpus sizes of neural networks (Brown et al. 2020; Kaplan et al. 2020). Increasing model sizes resulted in more interesting subjects of study, but also rendered causal interpretability significantly more difficult. Thus, a primary challenge in interpretability has been to balance the often contradictory goals of (i) obtaining a causal understanding of how and why models behave in a given manner, while also (ii) designing methods that are efficient and scalable.

Presently, there exist many subfields of interpretability that propose and apply causal methods to understand which model components contribute to an observed